Supplementary Material

Assessing Answer Span Detection of BERT-Based Architectures for Requirements Engineering QA Tasks: A Multi-Seed Robustness Study

This supplement reports the complete per-seed results for all ten model configurations (ten random seeds each). Aggregated results appear as Table 4 in the main manuscript.

Supplementary Table S1. Per-seed results for BERT

Seed	Ep	Min	EM	F1	ROUGE-L
13	3	17.93	0.4083	0.7095	0.7084
42	4	23.99	0.4333	0.7202	0.7202
73	4	24.33	0.4167	0.7516	0.7516
101	3	18.83	0.4333	0.7228	0.7218
123	3	17.92	0.3833	0.7381	0.7366
202	4	23.90	0.4167	0.7003	0.7003
333	3	20.30	0.3833	0.7178	0.7164
777	4	24.03	0.4167	0.7431	0.7431
1337	4	24.20	0.4583	0.7311	0.7311
2024	3	18.15	0.4167	0.7394	0.7377

Supplementary Table S2. Per-seed results for DistilBERT

Seed	Ep	Min	EM	F1	ROUGE-L
13	4	12.21	0.3833	0.6904	0.6881
42	5	15.29	0.3583	0.6873	0.6873
73	5	15.38	0.3917	0.7132	0.7132
101	5	15.44	0.3583	0.7173	0.7173
123	4	12.30	0.4000	0.6897	0.6897
202	5	15.20	0.4083	0.6894	0.6887
333	4	12.20	0.3833	0.7017	0.7017
777	5	15.19	0.4000	0.7208	0.7208
1337	5	15.19	0.3833	0.7153	0.7153
2024	5	28.48	0.3667	0.7031	0.7031

Supplementary Table S3. Per-seed results for BERT+LSTM 3L (frozen)

Seed	Ep	Min	EM	F1	ROUGE-L
13	13	12.29	0.1833	0.5205	0.5185
42	12	11.13	0.2083	0.5577	0.5557
73	9	8.64	0.2333	0.5790	0.5765
101	13	11.80	0.2250	0.5535	0.5522
123	13	11.40	0.2083	0.5428	0.5408
202	12	11.22	0.2333	0.5865	0.5835
333	13	11.25	0.2083	0.5515	0.5502
777	11	9.54	0.2250	0.5685	0.5656
1337	12	10.39	0.2083	0.5786	0.5749
2024	12	10.66	0.2667	0.5902	0.5878

Supplementary Table S4. Per-seed results for NeoBERT

Seed	Ep	Min	EM	F1	ROUGE-L

13	3	43.60	0.4583	0.7521	0.7521
42	3	43.75	0.4417	0.7380	0.7358
73	3	43.41	0.5000	0.7761	0.7761
101	3	44.38	0.4917	0.7794	0.7778
123	3	45.70	0.4667	0.7450	0.7450
202	3	44.17	0.5167	0.7906	0.7906
333	3	43.49	0.5250	0.8007	0.7993
777	3	44.17	0.4750	0.7478	0.7478
1337	3	43.38	0.4833	0.7753	0.7743
2024	3	43.41	0.4833	0.7722	0.7714

Supplementary Table S5. Per-seed results for SpanBERT

Seed	Ep	Min	EM	F1	ROUGE-L
13	4	32.72	0.4583	0.7244	0.7244
42	3	18.10	0.4083	0.7205	0.7205
73	4	24.06	0.4167	0.7099	0.7099
101	3	18.04	0.4750	0.7218	0.7218
123	4	24.06	0.4917	0.7859	0.7825
202	4	24.07	0.5250	0.7978	0.7967
333	4	24.07	0.4167	0.7195	0.7195
777	4	24.10	0.4667	0.7322	0.7307
1337	4	24.06	0.4667	0.7602	0.7591
2024	4	24.06	0.4417	0.7450	0.7440

Supplementary Table S6. Per-seed results for BERT+LSTM 2L (frozen)

Seed	Ep	Min	EM	F1	ROUGE-L
13	13	10.28	0.1417	0.4978	0.4954
42	12	9.50	0.1833	0.5435	0.5405
73	9	7.90	0.1917	0.5516	0.5484
101	12	9.50	0.2000	0.5664	0.5651
123	13	9.10	0.1667	0.4974	0.4974
202	12	8.03	0.1917	0.5363	0.5336
333	11	7.30	0.2333	0.5846	0.5807
777	11	7.34	0.1583	0.5162	0.5127
1337	9	6.75	0.2083	0.5549	0.5528
2024	12	9.87	0.2417	0.6000	0.5963

Supplementary Table S7. Per-seed results for BERT+LSTM 1L (frozen)

Seed	Ep	Min	EM	F1	ROUGE-L
13	12	6.71	0.1583	0.4921	0.4906
42	9	5.02	0.2083	0.5588	0.5567
73	9	5.00	0.2000	0.5555	0.5523
101	9	4.70	0.2250	0.5607	0.5567
123	10	5.78	0.2083	0.5281	0.5273
202	11	6.45	0.2083	0.5522	0.5501
333	11	5.71	0.2250	0.5774	0.5744
777	12	5.76	0.2167	0.5586	0.5534
1337	12	6.01	0.2333	0.5852	0.5828
2024	11	6.33	0.1917	0.5376	0.5344

Supplementary Table S8. Per-seed results for BERT+FF (frozen)

Seed	Ep	Min	EM	F1	ROUGE-L
13	20	6.35	0.1500	0.5417	0.5391
42	20	6.28	0.1583	0.5424	0.5409

73	20	6.49	0.1417	0.5358	0.5317
101	20	6.23	0.1583	0.5448	0.5431
123	20	6.21	0.1583	0.5398	0.5381
202	20	6.38	0.1417	0.5258	0.5229
333	20	6.46	0.1583	0.5366	0.5348
777	20	6.35	0.1500	0.5331	0.5306
1337	20	6.48	0.1583	0.5482	0.5465
2024	20	6.21	0.1583	0.5441	0.5423

Supplementary Table S9. Partial FT (last 2 BERT layers unfrozen + LSTM)

Seed	Ep	Min	EM	F1	ROUGE-L
13	6	29.21	0.3667	0.6942	0.6900
42	6	29.10	0.3833	0.7105	0.7094
73	6	29.34	0.3167	0.6353	0.6326
101	9	44.12	0.4500	0.7398	0.7398
123	5	24.52	0.3833	0.6888	0.6877
202	5	24.69	0.3333	0.6329	0.6311
333	5	24.51	0.4000	0.7377	0.7358
777	7	33.65	0.3833	0.7302	0.7283
1337	6	28.79	0.4500	0.7428	0.7409
2024	5	24.35	0.3583	0.6830	0.6812

Supplementary Table S10. Per-seed results for ModernBERT

Seed	Ep	Min	EM	F1	ROUGE-L
13	3	27.16	0.4333	0.7612	0.7602
42	3	27.20	0.4667	0.7731	0.7731
73	3	26.95	0.4667	0.7756	0.7746
101	3	26.78	0.4583	0.7456	0.7456
123	4	36.41	0.4250	0.7573	0.7573
202	4	36.55	0.4750	0.7662	0.7662
333	4	36.66	0.4500	0.7562	0.7554
777	3	26.77	0.4000	0.7034	0.7026
1337	4	35.68	0.4667	0.7649	0.7649
2024	3	26.78	0.4500	0.7445	0.7445

Supplementary Table S11. Full confirmatory statistics and equivalence-margin sensitivity (F1, 10 seeds)

Part A — Confirmatory pairwise tests (Holm correction applied within each hypothesis family).

Comparison	Family	Raw p	Holm p	Mean ΔF1	95% CI for mean ΔF1	Paired Cohen's d_z	Matched-pairs r_{rb}
NeoBERT vs BERT	H1	0.002	0.006	+0.040	[+0.019, +0.061]	1.37	1.00
ModernBERT vs BERT	H1	0.037	0.074	+0.027	[+0.006, +0.049]	0.92	0.75
SpanBERT vs BERT	H1	0.232	0.232	+0.014	[−0.013, +0.041]	0.38	0.45
Partial FT vs LSTM 3-layer	H3	0.002	0.004	+0.137	[+0.099, +0.174]	2.62	1.00
Partial FT vs BERT+FF (frozen)	H3	0.002	0.004	+0.160	[+0.133, +0.187]	4.28	1.00

Part B — Equivalence sensitivity (TOST via 90% CI of the mean difference; “Yes” = 90% CI within ±margin).

Comparison	Mean ΔF1	90% CI	±0.02	±0.03	±0.05
LSTM 1L vs 2L	+0.006	[−0.011, +0.023]	No	Yes	Yes
LSTM 1L vs 3L	−0.012	[−0.026, +0.001]	No	Yes	Yes
LSTM 2L vs 3L	−0.018	[−0.035, −0.001]	No	No	Yes
BERT+FF (frozen) vs LSTM 1L	−0.011	[−0.027, +0.004]	No	Yes	Yes
BERT+FF (frozen) vs LSTM 2L	−0.006	[−0.025, +0.014]	No	Yes	Yes
BERT+FF (frozen) vs LSTM 3L	−0.024	[−0.038, −0.010]	No	No	Yes

Part C — Unpaired sensitivity check (Mann–Whitney U on per-seed F1; does not assume seed pairing). Paired p from the Wilcoxon signed-rank test is shown for comparison.

Comparison	Paired p (Wilcoxon)	Unpaired p (Mann–Whitney)	Conclusion
NeoBERT vs. BERT	0.0020	0.0010	Significant under both
ModernBERT vs. BERT	0.0371	0.0036	Significant in the unpaired sensitivity analysis; not significant after H1-family Holm correction in the paired confirmatory analysis
SpanBERT vs. BERT	0.2324	0.3447	Not significant under either
Partial FT vs. LSTM 3-layer	0.0020	0.0002	Significant under both
Partial FT vs. BERT+FF (frozen)	0.0020	0.0002	Significant under both